SVM2

Thursday, March 14, 2024 9:03 PM

SVM - Logistic



Hard

no errors

Soft

errors

↓ Max Margin

↓ Min

Primal → Dual Primal

eqn 1 → eqn 2

} optimization

Same Task

Primal → w, b

Dual → α

↓

weight for each observation (Points)

of data point

Support Vectors
- Points that lie on hyperplane
- Margin (inside) (error)
- Misclassified (error)

Dual → α support

+1 Expension

-1 Middle

Predict → New value [sqm | Pool | Bed]

$$f(x_q) = \sum_{i=1}^{N} \alpha y_i (x_i^T x_q)$$

x — training data / support

> 1

$$y_q = \sum_{i=1}^{N} \alpha_i \cdot y_i (x_i^T \cdot x_q) \quad \geq 0$$

$y = 1$

$0 <$ $y = -1$

7:1 → Hard Margin
→ soft Margin

Dual Primal

Dual Primal

+

Kernel

$$\text{Max}_{\alpha} \sum_{i=1}^{N} \alpha_i - \frac{1}{2} \sum_{i=1}^{N} \sum_{j=1}^{N} \alpha_i \alpha_j y_i y_j \boxed{x_i^T x_j}$$

↓

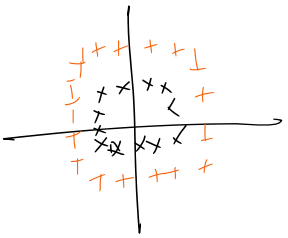
Dot product

Similarity

↓

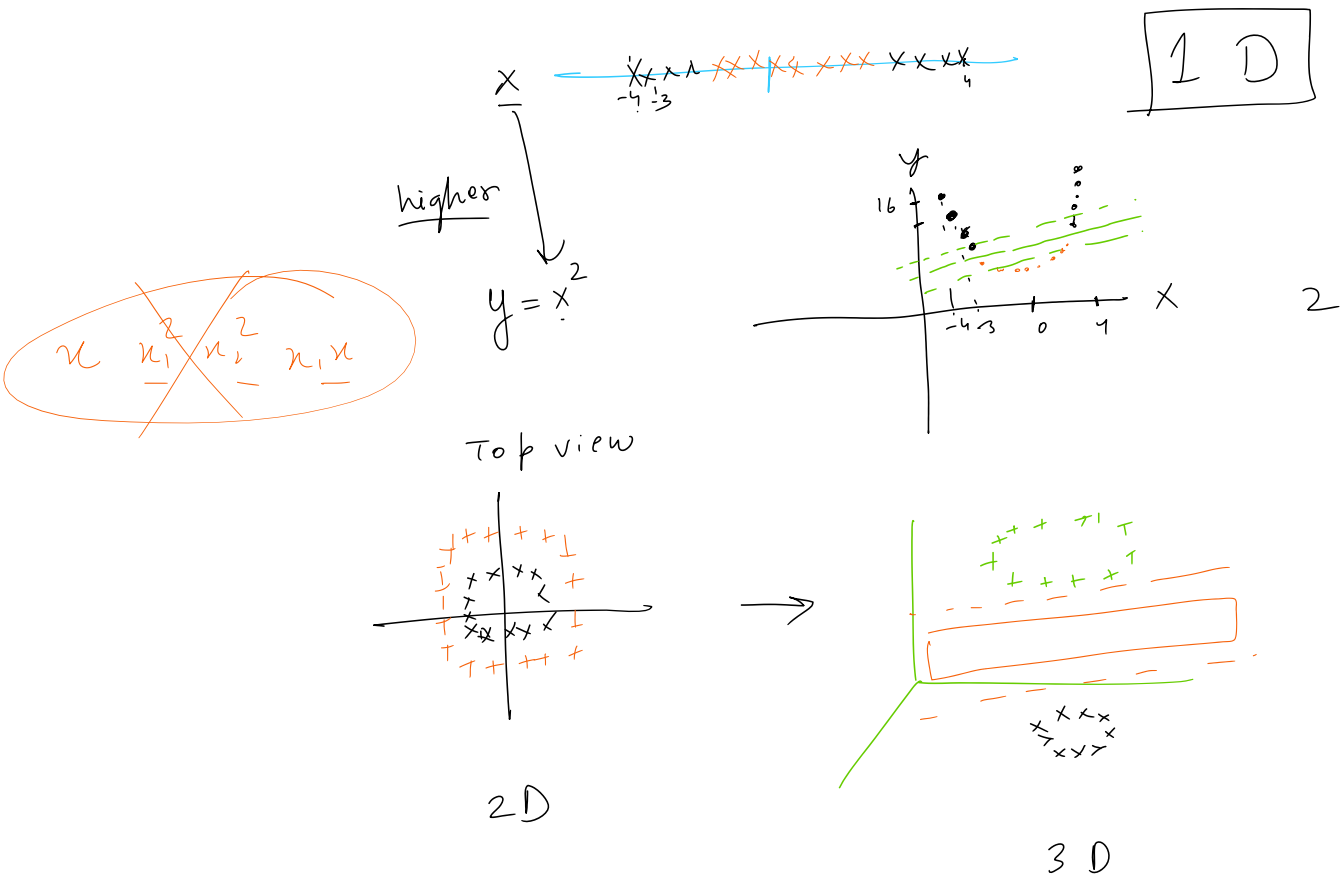
Kernel

$$\text{Cos}\theta = \frac{x^T y}{\|x\| \|y\|}$$



Kernel

↳ Non Linear



- Poly
- RBF

Code

Polynomial

$$(x_1\ x_2) = \left(x_1^T x_2 + c\right)^M \rightarrow \text{degree}$$

$M = 2$ Quadratic

$M = 3$ cubic

$$K_q(x_1, x_2) = (1 + x_1^T x_2)^2$$

DP$_1$ $\quad$ DP$_2$

$X_1 = [x_{11}, x_{12}]$ $\quad$ $X_2 = [x_{21}, x_{22}]$

$$\left(1 + \begin{bmatrix} x_{11} & x_{12} \end{bmatrix} \begin{bmatrix} x_{21} \\ x_{22} \end{bmatrix}\right)^2$$

Pair

$$\left(1 + x_{11}x_{21} + x_{12}x_{22}\right)^2$$

$$(a + b + c)^2$$

$$1 + x_{11}^2 x_{21}^2 + x_{12}^2 x_{22}^2 + 2x_{11}x_{21} + 2x_{12}x_{22} \ldots$$

$$[x_{11}, x_{12}] \hookrightarrow X_1 = \left[1, x_{11}^2, x_{12}^2, \sqrt{2}x_{11}, \sqrt{2}x_{12}, \sqrt{2}x_{11}x_{12}\right]$$

X_2

The input samples always will be

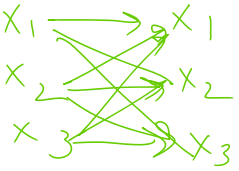
The input samples always will be in pair

$x_i \quad x_j$

$x_1 x_2 \quad \alpha > 0$

$x_2 x_3 \quad \alpha = 0$

$x_3 x_1 \quad \alpha = 0$



S.V

$= (1 + x_{11}x_{21} + x_{12}x_{22})^2 \qquad [\because (a+b+c)^2]$

$= 1 + x_{11}^2 x_{21}^2 + x_{12}^2 x_{22}^2 + 2x_{11}x_{21} + 2x_{12}x_{22} + 2x_{11}x_{21}x_{12}x_{22}$

Supposedly, we had two, 6D vectors x_1^1, x_2^1

s.t. $x_1^1 = [1, x_{11}^2, x_{12}^2, \sqrt{2}x_{11}^2, \sqrt{2}x_{12}^2, \sqrt{2}x_{11}x_{12}]$

$x_2^1 = [1, x_{21}^2, x_{22}^2, \sqrt{2}x_{21}^2, \sqrt{2}x_{22}^2, \sqrt{2}x_{21}x_{22}]$

If we perform dot product of x_1', x_2'

$$K(x_1, x_2) = (x_1'^T x_2')$$

This is same as what we got for **Quadratic Kernel**.

KNN → (n d)

↓

support vector

Ex y naa

O (n of suppo x d)

Test

Training

$\alpha = 0 \rightarrow 95\%$

$x_1 \quad x_2$

$n \quad n$

$O(n^2)$